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 2026 3 24

OSF DOI: 10.17605/OSF.IO/G9KJW CC BY 4.0

Abstract

We report four new geometric laws arising from the Sangbun Vector construction $AA12^s = AA6 + AA11$, defined on a line $y = ax + b$ with respect to the origin O . The laws concern: (1) a global balance between the projection component $|AA6|$ and the rotation component $|AA11|$; (2) a limit direction law in which $\angle AA12^s$ converges to $\arctan(a) \pm 45^\circ$; (3) a (Sangbun Slope Transformation) in which the limit slope of $AA12^s$ equals $(a - 1)/(a + 1)$, an intrinsic conversion induced by the Sangbun geometric construction; and (4) an oscillation mean recovery in which the average of the two oscillating limit angles recovers the original tangent direction. These results reveal a new geometric slope-conversion structure naturally induced by the Sangbun vector construction.

1. Background: (Sangbun Vector Construction)

The Sangbun vector $AA12^s$ is constructed geometrically from a point $A(x, ax + b)$ on the line $y = ax + b$, with respect to the origin $O(0, 0)$, through a sequence of projections, intersections, and rotations generating points A_1 through A_{12} .

The final vector is defined as:

$$\overrightarrow{AA12^s} = \overrightarrow{AA6} + \overrightarrow{AA11} \quad (1)$$

where $\overrightarrow{AA6}$ encodes the projection-vector structure and $\overrightarrow{AA11}$ encodes the rotation structure.

The branch invariant is:

$$@ = \frac{|OA_1|}{|AA_1|} \quad (2)$$

which is dimensionless and governs the three geometric branches ($@ < 1$, $@ = 1$, $@ > 1$).

2. Law 1: ((Balance Law)) $\Sigma|AA6| = \Sigma|AA11|$

Law 1 ((Balance Law)). *For uniformly sampled points on $y = ax + b$, the total magnitudes of the projection component $AA6$ and the rotation component $AA11$ satisfy a global balance:*

$$\boxed{\sum |AA6| \approx \sum |AA11| \quad (ratio \approx 1 : 1)} \quad (3)$$

2.1 Verification ($y = 3x + 5, x \in [-100, 100]$)

step	$\Sigma AA6 $	$\Sigma AA11 $	ratio	match
1.0	31954.51	31693.27	1.0082	✓
0.5	63595.69	63073.97	1.0083	✓
0.1	316794.53	314133.99	1.0085	✓

: () + ChatGPT + Gemini + Claude.

2.2 Geometric Meaning

Since $AA12^s = AA6 + AA11$, this balance reveals that the Sangbun vector distributes equally between its projection and rotation components, demonstrating an intrinsic self-consistency of the construction.

3. Law 2: ((Limit Direction Law)) ($\pm 45^\circ$)

Law 2 ((Limit Direction Law)). As $x \rightarrow \pm\infty$ along $y = ax + b$:

$$\lim_{x \rightarrow +\infty} \angle AA12^s = \arctan(a) - 45^\circ \quad (4)$$

$$\lim_{x \rightarrow -\infty} \angle AA12^s = \arctan(a) + 135^\circ \quad (5)$$

3.1 Verification

Line	$\arctan(a)$	$x \rightarrow +\infty$ diff.	$x \rightarrow -\infty$ diff.	match
$y = x + 3$	45.000°	-45.000°	$+135.003^\circ$	✓
$y = 2x + 3$	63.435°	-45.000°	$+135.001^\circ$	✓
$y = 3x + 5$	71.565°	-45.000°	$+135.001^\circ$	✓
$y = 5x + 3$	78.690°	-45.000°	$+135.000^\circ$	✓
$y = 10x + 3$	84.289°	-45.000°	$+135.000^\circ$	✓
$y = -3x + 5$	-71.565°	-45.000°	$+135.001^\circ$	✓

3.2 Geometric Meaning

As $x \rightarrow \infty$, $@ \rightarrow 0$. In this limit, $\overrightarrow{AA6}$ aligns with the tangent direction and $\overrightarrow{AA11}$ aligns with the normal direction. Their vector sum $AA12^s = AA6 + AA11$ converges to the bisector between tangent and normal — precisely 45° .

4. Law 3: ((Sangbun Slope Transformation))

Law 3 ((Sangbun Slope Transformation)). As $x \rightarrow +\infty$ along $y = ax + b$, the limiting slope of $AA12^s$ is:

$$\boxed{\lim_{x \rightarrow +\infty} \text{slope}(AA12^s) = \frac{a-1}{a+1}} \quad (6)$$

This reveals a new geometric slope-conversion structure naturally induced by the Sangbun vector construction. The input slope a is transformed into a new slope $(a - 1)/(a + 1)$ through the internal geometric process $A_1 \rightarrow A_{12}$.

4.1 Verification

slope a	computed	$(a - 1)/(a + 1)$	diff.	match
1.5	0.200000	0.200000	0.000000	✓
2	0.333333	0.333333	0.000000	✓
3	0.500000	0.500000	0.000000	✓
5	0.666667	0.666667	0.000000	✓
7	0.750000	0.750000	0.000000	✓
10	0.818182	0.818182	0.000000	✓

4.2 Properties of the Transformation

The mapping $f(a) = \frac{a - 1}{a + 1}$ satisfies:

$$\begin{aligned}
 f(1) &= 0 && \text{(unit slope maps to horizontal)} \\
 f(0) &= -1 && \text{(horizontal maps to slope } -1) \\
 f(\infty) &= 1 && \text{(vertical maps to slope } 1) \\
 f(-1) &= \infty && \text{(slope } -1 \text{ maps to vertical)}
 \end{aligned}$$

All real slopes are mapped into the interval $(-1, 1)$. This transformation arises entirely from the step-by-step geometry of A_1 through A_{12} — it is an intrinsic consequence of the Sangbun construction, not an externally imposed rule.

4.3 Connection to Law 2

The (Sangbun Slope Transformation) is fully consistent with the $\pm 45^\circ$ limit direction law (Law 2):

$$\tan(\arctan(a) - 45^\circ) = \frac{a - 1}{a + 1} \quad (7)$$

This confirms that Laws 2 and 3 are two expressions of the same underlying geometric property.

5. Law 4: (Oscillation and Mean Recovery)

Law 4 ((Oscillation Mean Recovery)). *For $x > 0$, the direction angle of AA_{12}^s oscillates between two values:*

$$\arctan(a) - 45^\circ \longleftrightarrow \arctan(a) + 45^\circ \quad (8)$$

Their mean recovers the original tangent direction:

$$\boxed{\frac{[\arctan(a) - 45^\circ] + [\arctan(a) + 45^\circ]}{2} = \arctan(a)} \quad (9)$$

5.1 Example ($y = 3x + 5$, $\arctan 3 = 71.565^\circ$)

value 1	value 2	difference	mean	$\arctan(3)$
26.565°	116.565°	90.000°	71.565°	$71.565^\circ \checkmark$

5.2 Geometric Meaning

The $\pm 45^\circ$ oscillation is a structured geometric alternation. Averaging the two limit directions recovers the original tangent direction exactly, showing that the Sangbun vector preserves tangent information in a symmetric, distributed form.

6. (Summary)

Law	Formula	Name
1	$\sum AA6 = \sum AA11 $	(Balance Law)
2	$\lim \angle AA12^s = \arctan(a) \pm 45^\circ$	(Limit Direction Law)
3	$\lim \text{slope}(AA12^s) = \frac{a-1}{a+1}$	(Sangbun Slope Transformation)
4	$\text{mean } \angle AA12^s = \arctan(a)$	(Oscillation Mean Recovery)

The Sangbun vector encodes an intrinsic slope-conversion structure and oscillates $\pm 45^\circ$ symmetrically around the tangent direction.

Reference

(Kim, Sangbok, 2026). *Sangbun Vector Geometry: A Geometric Construction Framework*. OSF . DOI: [10.17605/OSF.IO/G9KJW](https://doi.org/10.17605/OSF.IO/G9KJW). CC BY 4.0.